

Real Analysis Cheat Sheet

Field Axioms

A set $\mathbb{F}$ is a **field** if it has operations $+, \cdot$ satisfying:

- **Associativity:** $a+(b+c) = (a+b)+c$, $a(bc) = (ab)c$
- **Commutativity:** $a+b = b+a$, $ab = ba$
- **Identities:** $\exists 0, 1 \in \mathbb{F}$ s.t. $a+0 = a$, $a \cdot 1 = a$
- **Inverses:** $\forall a, \exists -a$ s.t. $a+(-a) = 0$. If $a \neq 0$, $\exists a^{-1}$ s.t. $aa^{-1} = 1$
- **Distributivity:** $a(b+c) = ab+ac$

Ordered Field Axioms

An **ordered field** $\mathbb{F}$ contains a positive subset $P \subset \mathbb{F}$ where for $a > b \iff a-b \in P$:

- **Trichotomy:** For any $a \in \mathbb{F}$, exactly one is true: $a \in P$, $-a \in P$, or $a = 0$.
- **Closure properties:** If $a, b \in P$, then $a+b \in P$ and $ab \in P$.
- **Transitivity:** $a < b$ and $b < c \implies a < c$.
- **Translation:** $a < b \implies a+c < b+c$.
- **Multiplication:** $a < b$ and $c > 0 \implies ac < bc$.

Construction of $\mathbb{R}$ via Dedekind Cuts

We can formally construct $\mathbb{R}$ from $\mathbb{Q}$ using **Dedekind cuts**. A cut is a subset $A \subset \mathbb{Q}$ such that:

- $A \neq \emptyset$ and $A \neq \mathbb{Q}$ (non-trivial).
- If $x \in A$ and $y < x$ for $y \in \mathbb{Q}$, then $y \in A$ (downward closed).
- If $x \in A$, then $\exists z \in A$ such that $x < z$ (no maximum element).

Properties & Operations:

- $\mathbb{R}$ is the set of all Dedekind cuts.
- $A \leq B \iff A \subseteq B$.
- $A+B = \{a+b \mid a \in A, b \in B\}$.
- In $\mathbb{R}$, every non-empty bounded set has a least upper bound (Completeness).

Completeness & Supremum Property

- **Upper Bound:** $M \in \mathbb{R}$ is an upper bound for S if $\forall s \in S, s \leq M$.
- **Supremum (LUB):** $s = \sup S$ if s is an upper bound, and for any upper bound M , $s \leq M$.
- **Axiom of Completeness:** Every non-empty subset $S \subset \mathbb{R}$ bounded above has a supremum.
- **Archimedean Property:** $\forall x, y \in \mathbb{R}$ with $x > 0$, $\exists n \in \mathbb{N}$ such that $nx > y$.
- **Density of $\mathbb{Q}$:** Between any two real numbers $a < b$, there exists $q \in \mathbb{Q}$ with $a < q < b$.

Classes of Real Numbers

- **Natural Numbers ($\mathbb{N}$):** Inductive set containing 1.
- **Rational ($\mathbb{Q}$) / Irrational:** $q \in \mathbb{Q} \iff q = m/n$ for $m \in \mathbb{Z}, n \in \mathbb{N}$. Reals not in $\mathbb{Q}$ are irrational (e.g., $\sqrt{2}$).
- **Archimedean Principle:** For any $h > 0$ and $x \in \mathbb{R}$, $\exists k \in \mathbb{Z}$ such that $(k-1)h \leq x < kh$.
- **Geometric Interpretation:** Points on a line. The sets (a, b) , $[a, b]$, $(a, b]$, $[a, b)$ are intervals.

Lemmas on Completeness

Three equivalent topological principles stemming from completeness:

- **Nested Interval Lemma (Cauchy–Cantor):** For any sequence of nested closed intervals $I_1 \supset I_2 \supset \dots$ with lengths $|I_n| \rightarrow 0$, there exists a unique point $c \in \bigcap_{n=1}^{\infty} I_n$.
- **Finite Covering Lemma (Borel–Lebesgue / Heine–Borel):** Every open cover of a closed and bounded interval $[a, b]$ admits a finite subcover.
- **Limit Point Lemma (Bolzano–Weierstrass):** Every bounded infinite subset of $\mathbb{R}$ has at least one limit point in $\mathbb{R}$.

Countable & Uncountable Sets

- **Equinumerous Sets:** $A \sim B$ if there exists a bijection $f: A \rightarrow B$.
- **Countable Set:** A set A is countable if $A \sim \mathbb{N}$.
- **Properties of Countable Sets:**
 - Any infinite subset of a countable set is countable.
 - The union of countably many countable sets is countable.
 - $\mathbb{Q}$ is countable.
- **Cardinality of the Continuum:** $\mathbb{R}$ is uncountable. In fact, any non-degenerate interval (like $[0, 1]$) is uncountable and has the cardinality of the continuum.
- **Cantor’s Theorem:** The power set 2^A has strictly greater cardinality than A .

Limits of Sequences

A sequence $\{x_n\}$ **converges** to a ($\lim_{n \rightarrow \infty} x_n = a$) if:

$$\forall \varepsilon > 0, \exists N \in \mathbb{N} \text{ s.t. } \forall n \geq N, |x_n - a| < \varepsilon$$

- **Cauchy Criterion:** Converges $\iff \forall \varepsilon > 0, \exists N$ s.t. $\forall m, n \geq N, |x_m - x_n| < \varepsilon$.
- **Monotone Convergence:** A monotone sequence converges $\iff$ it is bounded.
- **Bolzano-Weierstrass:** Every bounded sequence contains a convergent subsequence.

Limit Superior & Inferior:

- $\limsup x_n = \lim_{k \rightarrow \infty} (\sup_{n \geq k} x_n)$, $\liminf x_n = \lim_{k \rightarrow \infty} (\inf_{n \geq k} x_n)$.
- $x_n \rightarrow L \iff \limsup x_n = \liminf x_n = L$.
- The set of subsequential limits of a bounded sequence is closed and bounded. The largest is $\limsup x_n$.

Limits of Functions

$\lim_{x \rightarrow a} f(x) = L$ (using ε - δ definition):

$$\forall \varepsilon > 0, \exists \delta > 0 \text{ s.t. } 0 < |x - a| < \delta \implies |f(x) - L| < \varepsilon$$

- **Heine’s Definition (Sequences):** $\lim_{x \rightarrow a} f(x) = L \iff$ for every $\{x_n\} \rightarrow a$ ($x_n \neq a$), $f(x_n) \rightarrow L$.
- **Cauchy Criterion for Functions:** $\exists \lim_{x \rightarrow a} f(x) \iff \forall \varepsilon > 0, \exists \delta > 0$ s.t. $0 < |x' - a| < \delta$ and $0 < |x'' - a| < \delta \implies |f(x') - f(x'')| < \varepsilon$.
- **Limits over a Base ($\mathcal{B}$):** A more general framework for limits (e.g. $x \rightarrow \infty$, $x \searrow a$) using a filter base $\mathcal{B}$. $\lim_{\mathcal{B}} f(x) = L \iff \forall \varepsilon > 0, \exists B \in \mathcal{B}$ s.t. $x \in B \implies |f(x) - L| < \varepsilon$.

Properties of Limits

If $\lim x_n = A$ and $\lim y_n = B$, then:

- **Uniqueness & Boundedness:** Limits are unique. Convergent sequences are bounded.
- **Algebraic Operations:** $\lim(x_n \pm y_n) = A \pm B$; $\lim(x_n y_n) = AB$; $\lim(x_n/y_n) = A/B$ (if $y_n, B \neq 0$).
- **Order Properties:** If $x_n \leq y_n$ for all n , then $A \leq B$. (Strict inequality does *not* guarantee $A < B$).
- **Squeeze Theorem:** If $x_n \leq z_n \leq y_n$ and $A = B$, then $\lim z_n = A$.
- **One-Sided Limits:** $f(a \pm 0) = \lim_{x \rightarrow a^\pm} f(x)$. If f is monotone, 1-sided limits exist everywhere.

Elementary Facts About Series

A series $\sum_{n=1}^{\infty} x_n$ is a formal sum. The **partial sums** are $S_k = \sum_{n=1}^k x_n$.

- **Convergence:** The series converges to S if $\lim_{k \rightarrow \infty} S_k = S$.
- **Necessary Condition:** If $\sum x_n$ converges, then $\lim_{n \rightarrow \infty} x_n = 0$. (The converse is false, e.g., harmonic series $\sum 1/n$).
- **Cauchy Criterion for Series:** $\sum x_n$ converges $\iff \forall \varepsilon > 0, \exists N \in \mathbb{N}$ such that $\forall m > n \geq N, |x_{n+1} + \dots + x_m| < \varepsilon$.
- **The Number e :** $e = \sum_{n=0}^{\infty} \frac{1}{n!} = \lim_{n \rightarrow \infty} (1 + \frac{1}{n})^n$. It is irrational.

Series Convergence Tests

Let $\sum a_n$ and $\sum b_n$ be series.

- **Comparison Test:** If $0 \leq a_n \leq b_n$ for large n : (1) $\sum b_n$ converges $\implies \sum a_n$ converges. (2) $\sum a_n$ diverges $\implies \sum b_n$ diverges.
- **Limit Comparison Test:** If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = L > 0$, then $\sum a_n$ and $\sum b_n$ both converge or both diverge.
- **Cauchy Condensation Test:** If $a_n \geq 0$ is monotonically decreasing, then $\sum a_n$ converges $\iff \sum 2^n a_{2^n}$ converges.
- **p-Test:** $\sum \frac{1}{n^p}$ converges $\iff p > 1$ (proved via Cauchy Condensation or integral test).
- **Ratio (D’Alembert) Test:** $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$. Converges absolutely if $L < 1$. Diverges if $L > 1$. Inconclusive if $L = 1$.
- **Root (Cauchy) Test:** $L = \limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|}$. Converges absolutely if $L < 1$. Diverges if $L > 1$. Inconclusive if $L = 1$.
- **Alternating Series (Leibniz) Test:** If $a_n \geq 0$, monotonically decreasing $a_n \searrow 0$, then $\sum (-1)^{n+1} a_n$ converges.
- **Absolute Convergence:** $\sum |a_n|$ converges $\implies \sum a_n$ converges. (The converse is false; conditionally convergent if $\sum a_n$ converges but $\sum |a_n|$ diverges).

Continuous Functions: Definitions

Basic Definitions:

- A function $f: E \rightarrow \mathbb{R}$ is **continuous** at $a \in E$ if:
 $\forall \varepsilon > 0, \exists \delta > 0$ s.t. $\forall x \in E, |x - a| < \delta \implies |f(x) - f(a)| < \varepsilon$
- **Sequential criterion:** for any sequence $x_n \rightarrow a$, $f(x_n) \rightarrow f(a)$.

Points of Discontinuity:

- **Removable:** $\lim_{x \rightarrow a} f(x)$ exists but $\neq f(a)$.
- **First Kind (Jump):** Left limit $f(a-0)$ and right limit $f(a+0)$ exist but are unequal. Let $\omega(a) = |f(a+0) - f(a-0)|$ be the jump.
- **Second Kind:** At least one of the one-sided limits does not exist.

Continuous Functions: Properties

Local Properties:

- Continuity is preserved under $+$, $-$, $\cdot$, and $/$ (if denominator $\neq 0$).
- Composition of continuous functions is continuous.
- If f is continuous at a and $f(a) > 0$, then $f(x) > 0$ in some neighborhood of a .

Global Properties (on closed interval $[a, b]$):

- Weierstrass I (Boundedness):** f is bounded on $[a, b]$.
- Weierstrass II (Maximum):** f achieves its global maximum and minimum on $[a, b]$.
- Intermediate Value Theorem (Bolzano-Cauchy):** If $f(a)$ and $f(b)$ have opposite signs, $\exists c \in (a, b)$ s.t. $f(c) = 0$. By extension, f assumes all values between $f(a)$ and $f(b)$.
- Uniform Continuity (Cantor/Heine):** f is uniformly continuous on $[a, b]$.

Derivatives and Differentials

- Derivative:** $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$
- Differentiability:** $f(x+h) = f(x) + A \cdot h + o(h)$ as $h \rightarrow 0$, where $A = f'(x)$.
- Differential:** $df = f'(x)dx$, representing the linear part of the increment.
- Chain Rule:** $(g \circ f)'(x) = g'(f(x)) \cdot f'(x)$
- Inverse Function:** $(f^{-1})'(y) = \frac{1}{f'(x)}$ where $y = f(x)$ and $f'(x) \neq 0$.

Mean Value Theorems

- Fermat's Theorem:** If f has a local extremum at c and is differentiable there, then $f'(c) = 0$.
- Rolle's Theorem:** If $f \in C[a, b]$, differentiable on (a, b) , and $f(a) = f(b)$, then $\exists c \in (a, b)$ st $f'(c) = 0$.
- Lagrange MVT:** If $f \in C[a, b]$, differentiable on (a, b) , $\exists c \in (a, b)$ st $f'(c) = \frac{f(b) - f(a)}{b - a}$.
- Cauchy MVT:** If $f, g \in C[a, b]$, differentiable on (a, b) and $g'(x) \neq 0$, $\exists c \in (a, b)$ st $\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}$.
- L'Hôpital's Rule:** For $\frac{0}{0}$ or $\frac{\infty}{\infty}$, $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$.

Taylor's Formula

- Taylor's Formula:** $f(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k + R_n(x)$
- Peano Remainder:** $R_n(x) = o((x-a)^n)$ as $x \rightarrow a$.
- Lagrange Remainder:** $R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} (x-a)^{n+1}$ for some ξ between a and x .
- Cauchy Remainder:** $R_n(x) = \frac{f^{(n+1)}(\xi)}{n!} (x-\xi)^n (x-a)$.
- Maclaurin Series:** Taylor series centered at $a = 0$.

Riemann Integral & Properties

- Riemann Integral:** A function $f : [a, b] \rightarrow \mathbb{R}$ is Riemann integrable ($f \in \mathcal{R}[a, b]$) if there exists $I \in \mathbb{R}$ such that for every $\epsilon > 0$, there is a $\delta > 0$ where for any partition P with mesh $\|P\| < \delta$ and any sample points ξ_i , $|\sum_i f(\xi_i) \Delta x_i - I| < \epsilon$.
- Linearity:** If $f, g \in \mathcal{R}[a, b]$ and $\alpha, \beta \in \mathbb{R}$, then $\alpha f + \beta g \in \mathcal{R}[a, b]$ and

$$\int_a^b (\alpha f + \beta g) dx = \alpha \int_a^b f dx + \beta \int_a^b g dx$$

- Additivity:** For $a < c < b$, $f \in \mathcal{R}[a, b] \iff f \in \mathcal{R}[a, c]$ and $f \in \mathcal{R}[c, b]$, with

$$\int_a^b f dx = \int_a^c f dx + \int_c^b f dx$$

- Monotonicity:** If $f \leq g$ on $[a, b]$, then $\int_a^b f dx \leq \int_a^b g dx$.

Fundamental Theorems & Techniques

- First Fundamental Theorem of Calculus:** If f is continuous on $[a, b]$, then the function $F(x) = \int_a^x f(t) dt$ is differentiable on (a, b) and $F'(x) = f(x)$.
- Newton-Leibniz Formula (Second FTC):** If $f \in \mathcal{R}[a, b]$ and F is an antiderivative of f (i.e., $F' = f$), then

$$\int_a^b f(x) dx = F(b) - F(a)$$

- Integration by Parts:** If u, v are continuously differentiable on $[a, b]$, then

$$\int_a^b u(x) v'(x) dx = [u(x) v(x)]_a^b - \int_a^b u'(x) v(x) dx$$

- Substitution Rule:** If $\phi : [\alpha, \beta] \rightarrow [a, b]$ is continuously differentiable and f is continuous, then

$$\int_{\phi(\alpha)}^{\phi(\beta)} f(x) dx = \int_{\alpha}^{\beta} f(\phi(t)) \phi'(t) dt$$

Improper Integrals

- Improper Integrals (Unbounded Interval):** If f is integrable on $[a, R]$ for all $R > a$, the improper integral is defined as:

$$\int_a^{\infty} f(x) dx = \lim_{R \rightarrow \infty} \int_a^R f(x) dx$$

Converges if the limit exists and is finite.

- Improper Integrals (Unbounded Function):** If f is continuous on $[a, b)$ and $\lim_{x \rightarrow b^-} |f(x)| = \infty$, then:

$$\int_a^b f(x) dx = \lim_{\epsilon \rightarrow 0^+} \int_a^{b-\epsilon} f(x) dx$$

- Comparison Test:** If $0 \leq f(x) \leq g(x)$ for $x \geq a$:
 - If $\int_a^{\infty} g(x) dx$ converges, then $\int_a^{\infty} f(x) dx$ converges.
 - If $\int_a^{\infty} f(x) dx$ diverges, then $\int_a^{\infty} g(x) dx$ diverges.
- Absolute Convergence:** If $\int_a^{\infty} |f(x)| dx$ converges, then $\int_a^{\infty} f(x) dx$ converges absolutely.

Space $\mathbb{R}^m$... Topology

The space $\mathbb{R}^m$ is treated as a metric space with the standard Euclidean distance.

- Norm:** $\|x\| = \sqrt{x_1^2 + \dots + x_m^2}$. Satisfies $\|x\| \geq 0$, $\|\lambda x\| = |\lambda| \|x\|$, $\|x + y\| \leq \|x\| + \|y\|$ (Triangle Inequality).
- Distance:** $d(x, y) = \|x - y\|$.
- r -neighborhood:** $B_r(a) = \{x \in \mathbb{R}^m \mid \|x - a\| < r\}$.
- Open set:** $U \subset \mathbb{R}^m$ is open if $\forall x \in U$, $\exists r > 0$ s.t. $B_r(x) \subset U$.
- Closed set:** $F \subset \mathbb{R}^m$ is closed if its complement $\mathbb{R}^m \setminus F$ is open. Equivalently, F contains all its limit points.
- Compact set:** $K \subset \mathbb{R}^m$ is compact if every open cover has a finite subcover. *Heine-Borel:* K is compact $\iff K$ is closed and bounded.
- Bolzano-Weierstrass:** Every bounded sequence in $\mathbb{R}^m$ has a convergent subsequence.

Limits of Functions of Several Variables

Let $f : X \subset \mathbb{R}^m \rightarrow \mathbb{R}^n$ and a be a limit point of X .

- Limit:** $\lim_{x \rightarrow a} f(x) = L$ if $\forall \varepsilon > 0, \exists \delta > 0$ s.t. $0 < \|x - a\| < \delta \implies \|f(x) - L\| < \varepsilon$.
- Cauchy Criterion:** $\lim_{x \rightarrow a} f(x)$ exists $\iff \forall \varepsilon > 0, \exists \delta > 0$ s.t. $\forall x', x'' \in B_{\delta}(a) \cap X \setminus \{a\}$, $\|f(x') - f(x'')\| < \varepsilon$.
- Iterated Limits:** For $f(x, y)$, the iterated limits $\lim_{x \rightarrow a} (\lim_{y \rightarrow b} f(x, y))$ might not equal the multiple limit $\lim_{(x, y) \rightarrow (a, b)} f(x, y)$, and their existence does not imply the other.
- Properties:** Standard limit laws apply (linear combinations, products, quotients).

Continuity in $\mathbb{R}^n$

Let $f : X \subset \mathbb{R}^m \rightarrow \mathbb{R}^n$ be a function.

- Continuity at $a \in X$:** $\forall \varepsilon > 0, \exists \delta > 0$ s.t. $\|x - a\| < \delta \implies \|f(x) - f(a)\| < \varepsilon$. Equivalently, $\lim_{x \rightarrow a} f(x) = f(a)$.
- Global Continuity:** f is continuous on X if it is continuous at every point in X .
- Weierstrass Theorem:** If $K \subset \mathbb{R}^m$ is compact and $f : K \rightarrow \mathbb{R}$ is continuous, then $f(K)$ is compact (bounded) and achieves its maximum and minimum on K .
- Uniform Continuity:** f is uniformly continuous on X if $\forall \varepsilon > 0, \exists \delta > 0$ s.t. $\forall x_1, x_2 \in X$, $\|x_1 - x_2\| < \delta \implies \|f(x_1) - f(x_2)\| < \varepsilon$.
- Cantor's Theorem:** If f is continuous on a compact set K , then f is uniformly continuous on K .
- Connectedness:** If X is connected and f is continuous, $f(X)$ is connected (Intermediate Value Theorem equivalent).

Differentiability & Derivatives

- Partial Derivative:** Limit of f with respect to one variable, others held constant: $\frac{\partial f}{\partial x_i} = \lim_{h \rightarrow 0} \frac{f(x + h e_i) - f(x)}{h}$.
- Differentiability:** A function f is differentiable at x if $\Delta f = L(\Delta x) + o(\|\Delta x\|)$ as $\Delta x \rightarrow 0$, where L is a continuous linear operator (the differential $df(x)$).
- Jacobian Matrix:** The matrix representing the linear map $df(x)$, with entries $J_{ij} = \frac{\partial f_i}{\partial x_j}$.
- Chain Rule:** If f is diff. at x and g is diff. at $y = f(x)$, then $g \circ f$ is diff. at x with $J_{g \circ f}(x) = J_g(f(x)) J_f(x)$.
- Directional Derivative:** Rate of change along a vector v : $D_v f(x) = \lim_{t \rightarrow 0} \frac{f(x + tv) - f(x)}{t}$. If f is diff., $D_v f(x) = \nabla f(x) \cdot v$.
- Gradient:** Vector of partial derivatives $\nabla f = \left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n} \right)^T$. Points in the direction of steepest local ascent.

Taylor's Formula & Extrema

- Schwarz's Theorem:** If mixed partial derivatives f_{xy} and f_{yx} are continuous at a point, they are equal: $\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$.
- Taylor's Formula:** $f(x + h) = f(x) + df(x)(h) + \frac{1}{2!} d^2 f(x)(h, h) + \dots + \frac{1}{k!} d^k f(x)(h, \dots, h) + R_k(h)$, where remainder $R_k(h) = o(\|h\|^k)$. Up to second order using the Hessian matrix H : $f(x + h) \approx f(x) + \nabla f(x)^T h + \frac{1}{2} h^T H(x) h$.
- Extrema Necessary Condition:** For a differentiable function f , if a is a local extremum, then $\nabla f(a) = 0$ (critical or stationary point).
- Extrema Sufficient Condition (Hessian Matrix H):** Let $H(a) = \left[\frac{\partial^2 f}{\partial x_i \partial x_j} (a) \right]$ be the Hessian at a critical point a .
 - If $H(a)$ is positive definite, a is a **local minimum**.
 - If $H(a)$ is negative definite, a is a **local maximum**.
 - If $H(a)$ is indefinite, a is a **saddle point** (no extremum).

Inverse & Implicit Functions

- **Inverse Function Theorem:** Let $f : U \subset \mathbb{R}^n \rightarrow \mathbb{R}^n$ be C^k . If the Jacobian determinant $\det(J_f(a)) \neq 0$ at $a \in U$, then there exist neighborhoods V of a and W of $f(a)$ such that $f : V \rightarrow W$ is a C^k diffeomorphism (bijection with C^k inverse), and $J_{f^{-1}}(f(a)) = (J_f(a))^{-1}$.
- **Implicit Function Theorem:** Let $F : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^m$ be C^k . Suppose $F(a, b) = 0$ and the partial Jacobian matrix with respect to y , $\det(D_y F(a, b)) \neq 0$. Then near (a, b) , the solution to $F(x, y) = 0$ is locally given by a C^k function $y = \phi(x)$, and $D\phi(x) = -(D_y F)^{-1} D_x F$.
- **Functional Dependence:** Functions $y_1 = f_1(x), \dots, y_m = f_m(x)$ are functionally independent near a point iff the Jacobian matrix $[\frac{\partial f_i}{\partial x_j}]$ has maximum rank.
- **Constant Rank Theorem:** If J_f has constant rank k near a , then locally f is equivalent (via diffeomorphisms) to a linear projection/injection of rank k .

Extrema with Constraint & Morse Lemma

- **Constrained Extrema (Lagrange Multipliers):** To find local extrema of $f(x)$ subject to constraints $g_1(x) = 0, \dots, g_m(x) = 0$, define the Lagrangian $\mathcal{L} = f - \sum \lambda_i g_i$. A necessary condition at a (if gradients $\nabla g_i(a)$ are linearly independent) is $\nabla \mathcal{L}(a) = 0$, meaning $\nabla f(a)$ is a linear combination of $\nabla g_i(a)$.
- **Non-degenerate Critical Point:** A critical point a ($\nabla f(a) = 0$) where the Hessian matrix $H(a)$ is invertible (non-zero determinant).
- **Morse Lemma:** If a is a non-degenerate critical point of a C^3 function f , then there exists a local diffeomorphism $y = \phi(x)$ such that locally $f(y) = f(a) - y_1^2 - \dots - y_k^2 + y_{k+1}^2 + \dots + y_n^2$, where k is the index of the Hessian matrix $H(a)$ (number of negative eigenvalues).
- **Tangent & Normal Spaces:** For a surface $S = \{x : F(x) = 0\}$ where $\nabla F \neq 0$, the normal vector is ∇F , and the tangent space at $a \in S$ is $\{v \in \mathbb{R}^n : \nabla F(a) \cdot v = 0\}$.

Differential Calculus Properties

- **Darboux's Theorem:** If f is differentiable on $[a, b]$, then f' satisfies the intermediate value property (takes all values between $f'(a)$ and $f'(b)$).
- **Convexity & Jensen's Inequality:** f is convex on I if $f(tx + (1-t)y) \leq tf(x) + (1-t)f(y)$. For convex f , $f(\sum \lambda_i x_i) \leq \sum \lambda_i f(x_i)$ where $\sum \lambda_i = 1, \lambda_i \geq 0$.

Advanced Integration Properties

- **Lebesgue's Criterion:** A bounded function on $[a, b]$ is Riemann integrable $\iff$ its set of discontinuities has Lebesgue measure zero.
- **First Mean Value Theorem for Integrals:** If f is continuous, $\exists c \in [a, b]$ s.t. $\int_a^b f(x)g(x)dx = f(c) \int_a^b g(x)dx$ ($g \geq 0$).
- **Second Mean Value Theorem:** If f is monotonic and g integrable, $\exists c \in [a, b]$ s.t. $\int_a^b f(x)g(x)dx = f(a^+) \int_a^c g(x)dx + f(b^-) \int_c^b g(x)dx$.
- **Dirichlet / Abel Tests (Improper Integrals):** $\int f(x)g(x)dx$ converges if (Dirichlet) $F(x) = \int^x f$ is bounded and $g \searrow 0$, or (Abel) $\int f$ converges and g is monotone & bounded.

16.1 Pointwise & Uniform Convergence

- **Pointwise Convergence:** $f_n \rightarrow f$ on X if $\forall x \in X, \lim_{n \rightarrow \infty} f_n(x) = f(x)$.
- **Uniform Convergence ($f_n \rightrightarrows f$):** $\forall \varepsilon > 0, \exists N$ s.t. $\forall n \geq N, \forall x \in X, |f_n(x) - f(x)| < \varepsilon$. Equivalently, $\lim_{n \rightarrow \infty} \sup_{x \in X} |f_n(x) - f(x)| = 0$.
- **Cauchy Criterion:** $f_n \rightrightarrows f \iff \forall \varepsilon > 0, \exists N$ s.t. $\forall m, n \geq N, \sup_x |f_n(x) - f_m(x)| < \varepsilon$.
- **Dini's Theorem:** If (f_n) is a monotonic sequence of continuous functions converging point-wise to a continuous function f on a compact set K , then $f_n \rightrightarrows f$.

16.2 Uniform Convergence of Series

- **Partial Sums:** $\sum f_n(x) \rightrightarrows f(x)$ if $S_n(x) = \sum_{k=1}^n f_k(x) \rightrightarrows f(x)$.
- **Weierstrass M-Test:** If $|f_n(x)| \leq M_n$ on X and $\sum M_n < \infty$, then $\sum f_n(x)$ converges absolutely and uniformly on X .
- **Dirichlet's Test:** $\sum f_n(x)g_n(x)$ converges uniformly if the partial sums $F_n = \sum_{k=1}^n f_k$ are uniformly bounded, and $g_n(x) \rightrightarrows 0$ monotonically.
- **Abel's Test:** $\sum f_n(x)g_n(x)$ converges uniformly if $\sum f_n(x)$ converges uniformly and $g_n(x)$ is uniformly bounded and monotonic.

16.3 Properties of Uniform Convergence

- **Continuity:** If $f_n \rightrightarrows f$ on X and each f_n is continuous, then f is continuous.
- **Integration:** If $f_n \rightrightarrows f$ on $[a, b]$ and f_n are integrable, then f is integrable and $\lim_n \int_a^b f_n(x)dx = \int_a^b f(x)dx$. Same holds for series: $\int \sum f_n = \sum \int f_n$.
- **Differentiation:** If f_n are continuously differentiable on $[a, b]$, $\sum f_n(x_0)$ converges at some point x_0 , and $\sum f'_n(x)$ converges uniformly, then $\sum f_n(x)$ converges uniformly to a differentiable function f , and $f'(x) = \sum f'_n(x)$.

Function Spaces & Compactness

- **Arzelà-Ascoli Theorem:** A set $\mathcal{F} \subset C(K)$ (where K is compact) is relatively compact (every sequence has a uniformly convergent subsequence) $\iff \mathcal{F}$ is uniformly bounded and equicontinuous ($\forall \varepsilon > 0, \exists \delta > 0$ s.t. $\forall f \in \mathcal{F}, d(x, y) < \delta \implies |f(x) - f(y)| < \varepsilon$).
- **Stone-Weierstrass Theorem:** If a subalgebra $A \subset C(K; \mathbb{R})$ (K compact) separates points and contains constants, then A is dense in $C(K)$ under the uniform norm.